

Sydney Grammar School



TRIAL HSC EXAMINATION 1992

MATHEMATICS

3 UNIT

Time Allowed : 2 hours

13th August 1992

INSTRUCTIONS:

- * All questions may be attempted.
- * All questions are of equal value.
- * All necessary working must be shown.
- * Marks may not be awarded for careless or badly arranged work.
- * Approved calculators may be used.

SPECIAL INSTRUCTIONS:

- * Start each question on a new page.
- * Hand in each question separately.
- * Write your examination number on each page.

A table of standard integrals is included on the last page.

R.N.

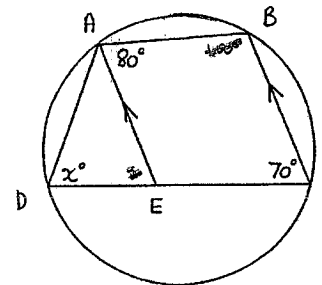
SGS HSC Trial 1992.....3 Unit Mathematics Form 6..... Page 2

QUESTION ONE (Start each question on a new page)

- (a) Solve for x the inequality $\frac{1}{x-1} < 5$.
- (b) (i) If $f(x) = x^2$ and $g(x) = \frac{1}{x}$, show algebraically that the graphs of $y = f(x)$ and $y = g(x)$ intersect at the point $(1, 1)$.
(ii) Find $f'(1)$ and $g'(1)$. Hence find the acute angle between the curves $y = x^2$ and $y = \frac{1}{x}$ at the point $(1, 1)$. (Give your answer correct to the nearest degree).
- (c) Use the substitution $u = x + 2$ to evaluate $\int_{-2}^0 x(x+2)^4 dx$.
- (d) (i) Find the exact value of $\cos [\sin^{-1}(-\frac{1}{2})]$.
(ii) Evaluate $\int_0^3 \frac{dx}{\sqrt{36-x^2}}$.
- (e) Find all value(s) of k for which the line $x - 2y + k = 0$ is a tangent to the circle $x^2 + y^2 = 4$.

QUESTION TWO (Start each question on a new page)

(a)



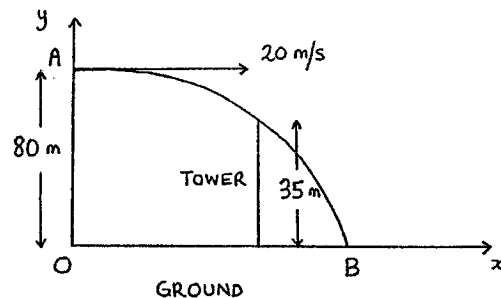
In the diagram above, $ABCD$ is a cyclic quadrilateral and E is a point on DC such that $AE \parallel BC$. $\angle BAE = 80^\circ$, $\angle BCD = 70^\circ$ and $\angle ADC = x^\circ$.

- (i) Copy this diagram onto your answer page.
- (ii) Find x , stating all reasoning.

(Exam continues next page...)

QUESTION TWO (Continued)

(b)



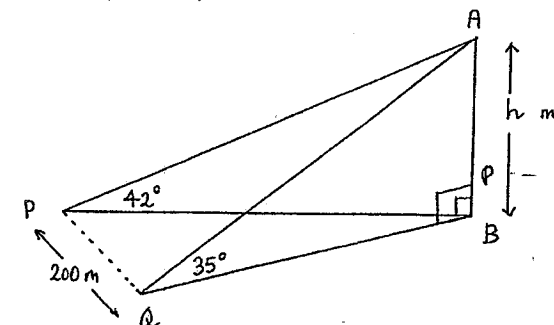
In the diagram above, an object is projected horizontally with a velocity of 20 m/s from a point A 80 metres above the ground and strikes the ground at point B.

- (i) If x and y are respectively the horizontal and vertical components of the object's displacement from O, derive expressions for x and y after t seconds. (Take O to be the origin and assume the acceleration due to gravity to be 10 m/s^2).
- (ii) Find the time taken for the object to reach the point B, and find the distance OB.
- (iii) If the object just clears a vertical tower of height 35 metres, find the distance from O to the base of the tower.

(Exam continues overleaf...)

QUESTION TWO (Continued)

(c)



The diagram above shows a point P, due west of B, from which the angle of elevation to the top of a tower AB height h metres, is 42° . From a point Q, bearing 196° from the tower, the angle of elevation to the top of the tower is 35° . The distance from P to Q is 200 metres.

(i) State the size of $\angle PBQ$.

(ii) Show that $h = \frac{200}{\sqrt{\cot^2 42^\circ + \cot^2 35^\circ - 2 \cot 35^\circ \cot 42^\circ \cos 74^\circ}}$.

(iii) Find h correct to 3 significant figures.

(Exam continues next page...)

QUESTION THREE (Start each question on a new page)

- (a) If α, β, γ are the roots of the equation:

$$2x^3 - 3x^2 + 5x + m = 0,$$

- (i) find the value of $\alpha + \beta + \gamma$ and $\alpha^2 + \beta^2 + \gamma^2$,

- (ii) find the value of m if $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -2$.

- (b) Use one application of Newton's method to find a three decimal place approximation to a root of the equation $e^x - 2x^2 = 0$. Use $x = 1.5$ as a first approximation to this root.

Copy and complete (correct to 2 decimal places) the following table of values for $P(x) = e^x - 2x^2$. Hence (i.e. without further calculation) explain why $x = 2.2$ would have been a less suitable first approximation for the root of $P(x) = 0$.

x	2.0	2.1	2.2	2.3	2.4
$P(x)$					

- (c) According to Newton's law of cooling, the rate at which a body cools in air is proportional to the difference between its temperature T° and the constant temperature P° of the surrounding air. This is expressed by the equation $T = P + Ae^{kt}$, where A, k are constants and t is the time in hours. The temperature of the air surrounding a heated body is 15°C , and the body cools from 90°C to 65°C in 3 hours.

- (i) Find the temperature of the body after a further 2 hours. (Answer to the nearest degree). Also find (in $^\circ\text{C}$ per hour) the body's rate of cooling at that time.

- (ii) Find (correct to the nearest minute) the time taken for the body to cool to 30°C from its original temperature.

(Exam continues overleaf...)

QUESTION FOUR (Start each question on a new page)

- (a) If $\sin \theta = \frac{5}{13}$, $0 < \theta < \frac{\pi}{2}$, find exact values for:

(i) $\sec \theta$,

(ii) $\cos \frac{\theta}{2}$.

- (b) P is the point $(4t, 2t^2)$ on the parabola $x^2 = 8y$.

- (i) Show that the equation of the normal to the parabola at P is $x + ty = 2t^3 + 4t$.

- (ii) If the normal at P cuts the y axis at point A , show that the coordinates of A are $(0, 2t^2 + 4)$.

- (iii) If R is the mid point of AP , show that the locus of R is a parabola. Find the vertex and focus of the parabola.

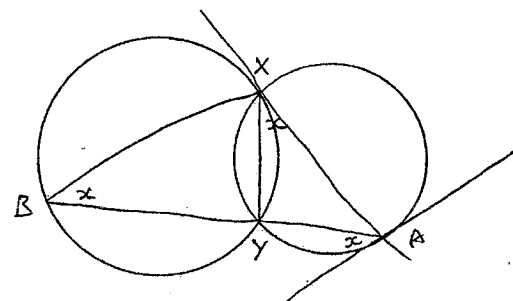
- (c) Use mathematical induction to prove that $7^n > 4^n + 5^n$ for all integers $n \geq 2$.

QUESTION FIVE (Start each question on a new page)

- (a) Solve the following equation over $0 \leq x \leq 2\pi$:

$$\cos 2x - 3 \sin x - 2 = 0.$$

- (b)



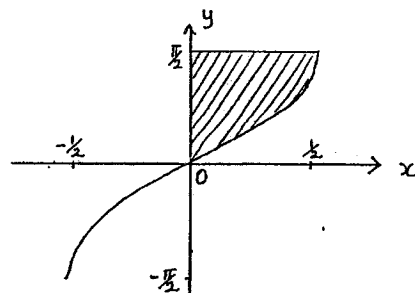
In the diagram above, two circles intersect at X and Y . The tangent at X to the larger circle cuts the smaller circle at A , and AY produced cuts the larger circle at B .

- (i) Draw a diagram showing all the given information.
(ii) The tangent to the smaller circle at A is drawn. Prove that this tangent is parallel to BX .

(Exam continues next page...)

QUESTION FIVE (Continued)

(c)



The diagram above shows the graph of $y = \sin^{-1}(2x)$. The shaded area is that bounded by the curve, the y axis and the line $y = \frac{\pi}{2}$.

- (i) Find this shaded area.
- (ii) If this area is rotated about the y axis, find the volume of the solid thus formed.

(Exam continues overleaf...)

QUESTION SIX (Start each question on a new page)

- (a) In the expansion of $(5 + 3x)^{20}$, in ascending powers of x , A_r is the coefficient of x^{r-1} in the term U_r , and A_{r+1} is the coefficient of x^r in the term U_{r+1} .

- (i) Use the Binomial Theorem to write expressions for A_{r+1} and A_r . Hence show that:

$$\frac{A_{r+1}}{A_r} = \frac{63 - 3r}{5r}.$$

- (ii) Hence find the greatest coefficient in the expansion of $(5 + 3x)^{20}$. (Leave your answer in the form ${}^{20}C_r 3^a 5^b$.)

- (b) On Wednesday morning at 5 a.m. an aeroplane crashes into a harbour. The rescue team and its equipment are most effective when the depth of water in the harbour is no more than 7 metres. At low tide the water is 5 metres deep, and at high tide the depth is 10 metres. Low tide occurs at 4 a.m. and high tide at 10.15 a.m. Assume that the movement of the tides is *SHM*.

- (i) State the period and amplitude of the motion.
- (ii) If the deadline for the rescue operation is 6 p.m. on Wednesday evening, find the periods of time between 5 a.m. and 6 p.m. during which the rescue team can work most effectively.

RN

Q. 1. (a) $\frac{1}{x-1} < 5$.

$(x-1) < 5(x-1)^2$

$5(x-1)^2 - (x-1) > 0$ ✓

$[x-1](5x-5-x+1) > 0$

$(x-1)(4x-4) > 0$

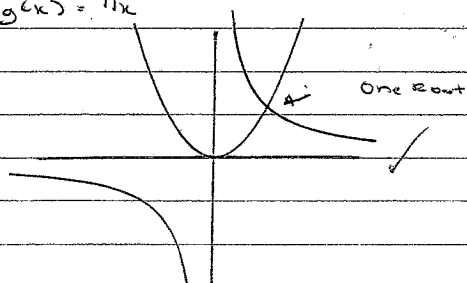
$4(x-1)^2 > 0$ ✓

$x < 1 \vee x > 1$

All real $x \neq 1$.

(b) (i) $f(x) = x^2$

$g(x) = 1/x$



$x^2 = 1/x$

$x^3 = 1$

$x = 1$

$(1, 1)$ ✓

(ii) $f'(x) = 2x$

$f'(1) = 2$ ✓

$g'(x) = -x^{-2}$

$g'(1) = -1/1$

$= -1$ ✓

$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{2 - (-1)}{1 - 2} \right|$

$\theta = \tan^{-1}(3)$

$= 71^\circ 34'$ ✓

(c) $\int_{-2}^0 x(x+2)^4 \cdot dx$

Let $u = x+2$
 $du = dx$

At $x=0, u=2$

$u=-2, u=0$ ✓

$I = \int_0^2 (u-2)u^4 \cdot du$

$= \int_0^2 u^5 - 2u^4 \cdot du$ ✓

$= \left[\frac{u^6}{6} - \frac{2u^5}{5} \right]_0^2$

$= \frac{16}{3}$

72/72

Excellent!

Q. 1. (a)

(a) (i) $\cos(\sin^{-1}(-1/2))$

$= \cos(-\pi/6)$ ✓

$= \sqrt{3}/2$

(ii)

$\int_0^{\pi/2} \frac{dx}{\sqrt{x^2 - k^2}}$

$= \sin^{-1} \left(\frac{x}{k} \right) \Big|_0^{\pi/2}$

$= \sin^{-1}(1/k)$ ✓

$= \pi/6$ ✓

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(c)

$x - 2y + k = 0$

$x = 2y - k$

$x^2 = (2y - k)^2$

$\therefore (2y - k)^2 + y^2 = 4$ ✓

$4y^2 - 4ky + k^2 + y^2 - 4 = 0$

$5y^2 - 4ky + (k^2 - 4) = 0$

$\Delta = 0$ ✓

$16k^2 - 20(k^2 - 4) = 0$

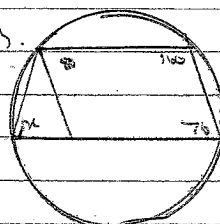
$16k^2 - 20k^2 + 80 = 0$

$-4k^2 + 80 = 0$

$20 = k^2$ ✓

$k = \pm 2\sqrt{5}$

2. (a)



(i) $x = 80^\circ$

(App. angle at center quad arc $\rightarrow \pi$)

(b)

$\ddot{x} = 0$

$\dot{x} = \int \ddot{x} \cdot dt$

$= C_1$

At $t=0, \dot{x} = \sqrt{v \cos \theta}$ ✓

$\therefore \dot{x} = \sqrt{v \cos \theta}$

$x = \int \dot{x} \cdot dt$

$= \sqrt{v \cos \theta} \cdot t + C_2$

At $t=0, x=0$

$x = \sqrt{v \cos \theta} \cdot t$ ✓

$x = 20t$

$\ddot{y} = -g$

$\dot{y} = \int \ddot{y} \cdot dt$

$= -gt + C_3$

At $t=0, \dot{y} = \sqrt{v \sin \theta}$

$\dot{y} = -gt + \sqrt{v \sin \theta}$

$y = \int \dot{y} \cdot dt$

$= -\frac{gt^2}{2} + \sqrt{v \sin \theta} \cdot t + C_4$

At $t=0, y=0$

$\therefore y = -\frac{gt^2}{2} + \sqrt{v \sin \theta} \cdot t + 80$

$y = -5t^2 + 80$

(ii). $y = -5t^2 + 80$

$y = 0, \quad 80 = 5t^2$

$16 = t^2$

$t = 4 \text{ s.}$

$t = 4 \therefore x = 80$

$\therefore OB = 80 \text{ metres.}$

(iii). $x = 20t$

$t = x/20$

$y = -5t^2 + 80$

$y = \frac{-5x^2}{400} + 80$

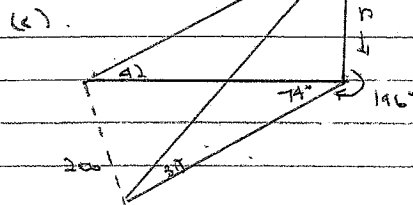
$y = 35$

$35 = \frac{-5x^2}{400} + 80$

$14000 = -5x^2 + 32000$

$3600 = x^2$

$x = 60 \text{ metres.}$



(i) $\hat{P}BQ = 74^\circ$

(ii) $\hat{Q}B = \tan 35$

$QB = h \cot 35$

$PB = h \cot 42$

$\therefore \triangle PQB$

$PQ^2 = PB^2 + QB^2 - 2PB \cdot QB \cos \hat{P}BQ$

$200^2 = h^2 \cot^2 35 + h^2 \cot^2 42 - 2h^2 \cot 35 \cot 42 \cos 74$

$200^2 = h^2 (\cot^2 35 + \cot^2 42 - 2 \cot 35 \cot 42 \cos 74)$

$h^2 = \frac{200^2}{\cot^2 35 + \cot^2 42 - 2 \cot 35 \cot 42 \cos 74}$

$(\cot^2 35 + \cot^2 42 - 2 \cot 35 \cot 42 \cos 74)$

$h = 200$

(iii). $h = 129 \text{ m.}$

3.(a). $2x^3 - 3x^2 + 5x + m = 0$

$\alpha + \beta + \gamma = 3/2$

$(\alpha^2 + \beta^2 + \gamma^2) = (\sum \alpha)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$= \frac{9}{4} - 2(5/2)$

$= 9/4 - 5$

$= -11/4$

(ii). $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -2$

$\frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = -2$

$\frac{5/2}{-m/4} = -2$

$5/2 = m$

$m = 5/2$

(b). $e^x - 2x^2 = 0$

$f(x) = e^x - 4x$

$x_1 = 1.5 - \frac{f(1.5)}{f'(1.5)}$

$= 1.5 - (-0.018)$

-1.518

$\frac{x+2}{P(x)} = \frac{1.49}{P(x)} = \frac{1.49}{e^x - 2x^2}$

x	2	2.1	2.2	2.3	2.4
P(x)	-0.6	-0.65	0.65	0.6	-0.4

$P(2.2) = \text{greatest value}$

$\therefore \text{ T.P. at } x = 2.2$

$\therefore \text{ Less suitable at tangent to}$

(c). $T = P + Ae^{kt}$

$T = 15 + Ae^{kt}$

$\Delta t = 0, T = 90$

$90 = 15 + A$

$A = 75$

$\Delta t = 3, T = 65$

$65 = 15 + 75e^{3k}$

$2/3 = e^{3k}$

$\ln(2/3) = k$

(i). $t = 5$.

$$T = 15 + 75e^{\frac{\ln(2/3)t}{3}}$$

$$= 53^\circ$$

(ii). $30 = 15 + 75e^{\frac{\ln(2/3)t}{3}}$

$$15/75 = e^{\frac{\ln(2/3)t}{3}}$$

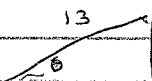
$$\ln(1/5) = \ln(2/3)t$$

$$\frac{3 \ln(1/5)}{\ln(2/3)} = t$$

$$t = 11.9$$

∴ 12 hrs.

4. (a). $\sin \theta = 5/13$.



(i) $\sec \theta = 13/5$

(ii) $\cos \theta/2$.

$$2 \sin \theta/2 \cos \theta/2 = 5/13$$

$$L + t = \tan \theta/2$$

$$\sin \theta = \frac{2t}{1+t^2} = 5/13$$

$$26t = 5 + 5t^2$$

$$5t^2 - 26t + 5 = 0$$

$$t = 5, 1/5$$

$$\tan \theta/2 = 5$$

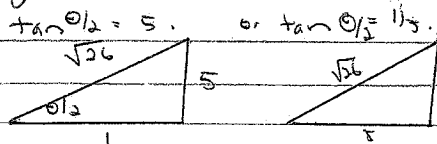
Quicker to use

$$\cos \theta = 2 \cos^2 \theta/2 - 1$$

$$\frac{12}{13} = 2 \cos^2 \theta/2 - 1$$

$$\frac{25}{16} = \cos^2 \theta/2$$

$$\therefore \cos \theta/2 = \frac{5}{\sqrt{16}} \text{ since } \theta/2 \in [0, \pi/2]$$



$$\therefore \cos \theta/2 = 1/\sqrt{26} \text{ or } 5/\sqrt{26}$$

$$\text{But } 0 < \theta/2 \leq \pi/2$$

$$\therefore \cos \theta/2 = \frac{5}{\sqrt{26}}$$

(b) (i) $P(4t, 2t^2)$

$$\frac{x^2}{8} = y$$

$$\frac{dy}{dx} = \frac{2x}{8}$$

$$at P(4t, 2t^2)$$

$$m = \frac{2t}{4}$$

$$= t$$

$$m_2 = -\frac{1}{t}$$

$$y - 2t^2 = -\frac{1}{t}(x - 4t)$$

$$ty - 2t^3 = -x + 4t$$

$$\underline{x + ty = 2t^3 + 4t}$$

(ii). A, when $x = 0$.

$$y = 2t^3 + 4t$$

$$y = 2t^2 + 4$$

$$\therefore A(0, 2t^2 + 4)$$

(iii) $P(4t, 2t^2) \quad A(0, 2t^2 + 4)$

$$\therefore x = 2t, y = 5t^2 + 2$$

$$t = x/2, y = 5(\frac{x}{2})^2 + 2$$

$$y = \frac{x^2}{2} + 2$$

$$2y = x^2 + 4$$

$$2y + 4 = x^2$$

$$4(\frac{y}{2} - 1) = x^2$$

$$\text{Vertex: } (0, 2)$$

ac1

$$\text{Focus: } (0, 1)$$

(c). $T^n > 4^n + 5^n$

$$L + n = 2$$

$$\text{LHS: } T^2$$

$$\text{RHS: } 41$$

$$= 41$$

$$\therefore \text{LHS} > \text{RHS}$$

Assume true for $n = k$.

$$T^k > 4^k + 5^k$$

Prove true for $n = k+1$

$$\text{LHS} = T^{k+1} > 7(4^k + 5^k)$$

$$7^k - 4^k - 5^k > 0.$$

DTF, true $n=k+1$

$$7^{k+1} - 4^{k+1} - 5^{k+1} > 0. \quad \checkmark$$

$$LHS = 7(7^k - 4^k - 5^k) + 3 \cdot 4^k + 0 \cdot 5^k$$

$$= 7 \cdot 7^k - 7 \cdot 4^k - 7 \cdot 5^k + 3 \cdot 4^k = 7^{k+1} - 4^{k+1} - 7 \cdot 5^k$$

Statement:

$$5. (a). \cos 2x - 3 \sin x - 2 = 0.$$

$$(1 - \sin^2 x) - 3 \sin x - 2 = 0.$$

$$-2 \sin^2 x - 3 \sin x - 1 = 0.$$

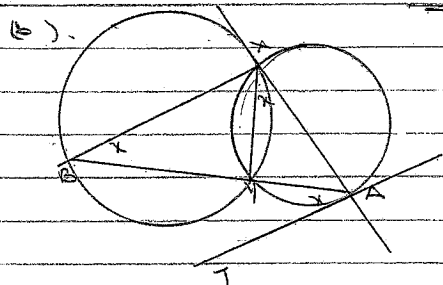
$$2 \sin^2 x + 3 \sin x + 1 = 0.$$

$$\sin x = \frac{-3 \pm \sqrt{9 - 8}}{4}$$

$$= \frac{-3 \pm 1}{4}$$

$$= -\frac{1}{2} \text{ or } -1$$

$$x = 11\pi/6, \frac{7\pi}{6}, \frac{3\pi}{2}$$



$$L + \hat{YAT} = x.$$

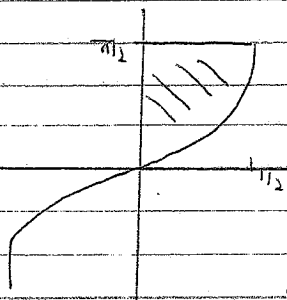
$$\hat{YKA} = x \quad (\text{L in the alternate segments})$$

$$\hat{KBA} = x \quad (\text{...})$$

$$\therefore \hat{XBA} = \hat{TAB}$$

$$\therefore BX \parallel TA \quad (\text{alternate angles are equal}).$$

(2).



$$y = \sin^{-1}(2x)$$

$$\sin y = 2x$$

$$\text{Area} = \int_{-1/2}^{1/2} x \cdot dy$$

$$= \frac{1}{2} \int_{-\pi/2}^{\pi/2} \sin y \cdot dy$$

$$= \frac{1}{2} [-\cos y]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{2}$$

(ii).

$$\text{Volume} = \pi \int_0^{\pi/2} x^2 \cdot dy$$

$$= \frac{\pi}{4} \int_0^{\pi/2} \sin^2 y \cdot dy$$

$$= \frac{\pi}{4} \int_0^{\pi/2} (1 - \cos 2y) \cdot dy$$

$$= \frac{\pi}{4} [y - \frac{1}{2} \sin 2y]_0^{\pi/2}$$

$$= \frac{\pi^2}{16}$$

$$\cos 2x = 1 - 2 \sin^2 x$$

$$1 - \cos 2x = 2 \sin^2 x$$

$$6. (a). (5+3x)^{20}.$$

$$\frac{A_{r+1}}{A_r} = \frac{{}^{20}C_r (5)^{20-r} (3x)^r}{{}^{20}C_{r-1} (5)^{20-r+1} (3x)^{r-1}}$$

$$= \frac{20!}{(20-r)! r!} \times \frac{(5)^{20-r} (3x)^r}{(5)^{20-r+1} (3x)^{r-1}}$$

$$= \frac{20!}{(20-r)! (r-1)!} \times \frac{(5)^{20-r} (3x)^r}{(5)^{20-r+1} (3x)^{r-1}}$$

$$= \frac{(20-r)! (r-1)!}{(20-r)! (r-1)!} \times 3$$

$$= \frac{(20-r)!}{(20-r)!} \times 3$$

$$= 3$$

$$5r.$$

$$(ii). 63-3r > 1$$

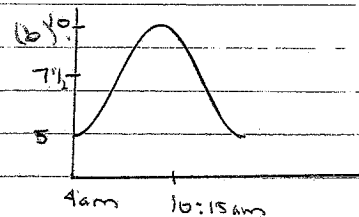
$$5r$$

$$63-3r > 5r$$

$$63 > 8r$$

$$r < 7.875$$

$$\therefore \frac{{}^{20}C_7 (3)^{13} (5)^7}{\dots}$$



$$(i) T = \frac{2\pi}{\omega} = \frac{2\pi}{4\pi/25}$$

$$25\pi = 4\pi$$

$$n = 4\pi/25$$

$$\text{Amplitude } 2\pi/25.$$

$$(ii). x = 7\frac{1}{2} - 2\frac{1}{2} \cos\left(\frac{4\pi}{25}t\right)$$

$$2x = 15 - 5 \cos\left(\frac{4\pi}{25}t\right)$$

$$x = 7.5$$

$$14 = 15 - 5 \cos\left(\frac{4\pi}{25}t\right)$$

$$1/5 = \cos\left(\frac{4\pi}{25}t\right)$$

$$1/36 = 4\pi/25 t$$

$$t = 2.72$$

$$\text{or } 4 \cdot 9137 = 4\pi/25 t.$$

$$t = 9.775$$

$$\therefore 8:06 \text{ am.}$$

$$2:47 \text{ pm}$$

$$\therefore \text{Between } 5 \text{ am} - 8:06 \text{ am}$$

$$2:47 \text{ pm} - 6 \text{ pm.}$$